

EXP NO:	6. Implement Linux Privilege Escalation Checker.
DATE:	

AIM:

ALGORITHM:

231401043

PROGRAM:

```
import subprocess

def run(command):

    try:

        result = subprocess.run(

            command,

            shell=True,

            text=True,

            capture_output=True

        )

        return result.stdout.strip()

    except Exception:

        return "Unable to execute command"


print("Linux Privilege Escalation Checker")

print("-" * 40)

print("\n1. Current User")

print(run("whoami"))

print(run("id"))


print("\n2. Sudo Permissions")

sudo = run("sudo -n -l 2>/dev/null")

if sudo:

    print(sudo)

else:
```

```
print("No passwordless sudo permission found.")

print("\n3. SUID Files")

suid = run(
    "find /usr/bin /usr/sbin /bin /sbin "
    "-perm -4000 -type f 2>/dev/null"
)

if suid:
    print(suid)
else:
    print("No SUID files found.")

print("\n4. World-Writable Files")

writable = run(
    "find /etc /usr /opt "
    "-type f -perm -0002 2>/dev/null | head -20"
)

if writable:
    print(writable)
else:
    print("No world-writable files found.")

print("\n5. Sensitive File Permissions")

print(run("ls -l /etc/passwd /etc/shadow 2>/dev/null"))

print("\n6. Writable Cron Files")

cron = run(
    "find /etc/cron* -type f -writable 2>/dev/null"
)
```

if cron:

print(cron)

else:

print("No writable cron files found.")

print("\nScan Completed Successfully.")

OUTPUT:

```
Linux Privilege Escalation Checker
-----
1. Current User
root
uid=0(root) gid=0(root) groups=0(root)

2. Sudo Permissions
Matching Defaults entries for root on DESKTOP-18EADPO:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty

User root may run the following commands on DESKTOP-18EADPO:
    (ALL : ALL) ALL

3. SUID Files
/usr/bin/chsh
/usr/bin/fusermount3
/usr/bin/chfn
/usr/bin/sudo
/usr/bin/newgrp
/usr/bin/su
/usr/bin/umount
/usr/bin/mount
/usr/bin/gpasswd
/usr/bin/passwd

4. World-Writable Files
No world-writable files found.

5. Sensitive File Permissions
-rw-r--r-- 1 root root 1380 Jan 6 2025 /etc/passwd
-rw-r----- 1 root shadow 702 Jan 6 2025 /etc/shadow

6. Writable Cron Files
/etc/cron.d/e2scrub_all
/etc/cron.d/.placeholder
/etc/cron.daily/apport
/etc/cron.daily/dpkg
/etc/cron.daily/logrotate
/etc/cron.daily/apt-compat
/etc/cron.daily/man-db
/etc/cron.daily/.placeholder
/etc/cron.hourly/.placeholder
/etc/cron.monthly/.placeholder
/etc/cron.weekly/man-db
/etc/cron.weekly/.placeholder
/etc/cron.yearly/.placeholder
/etc/crontab

Scan Completed Successfully.
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DESKTOP-18EADPO: /usr/bin/sudo -l -U root -H 2025-01-06 13:05:34
```

RESULT: